

Fifty Years of YM on $\mathbb{R}^4$: Where Every Approach Smuggles a Scale

Cal (Claude 4.7) — YM-11 deliverable, Hodge-template YM closure sprint

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Fifty Years of YM on $\mathbb{R}^4$: Where Every Approach Smuggles a Scale

Purpose

The Clay Yang-Mills problem requires a quantum YM theory on $\mathbb{R}^4$ with positive mass gap satisfying the Wightman axioms. After half a century of investigation, no approach has delivered all three simultaneously on $\mathbb{R}^4$.

This catalog tabulates every major school's attempt, identifying:

1. Where each approach smuggles a length scale — the mechanism by which a scale enters that is not derivable from the Wightman axioms alone.
2. Which Wightman axioms cannot be verified — specifically W3 (spectrum condition with mass gap) and W4 (local commutativity / microcausality) in the continuum limit on $\mathbb{R}^4$.

The pattern is structural: every approach succeeds in some restricted setting (finite volume, lower dimension, lattice, AdS background) and fails in the same place when the restriction is removed. This is the empirical content of the Curvature/Boundary Necessity claim formalized in Paper YM-C, Section 2.

The Catalog

A. Constructive QFT

A.1 Glimm-Jaffe construction (1968-1987)

- Achievement: Rigorous Wightman QFT for ϕ_2^4 (1968-1972) and ϕ_3^4 (1974-1987). Schwinger functions verified to satisfy Osterwalder-Schrader axioms; Wightman reconstruction yields all five axioms.
- Dimension: 2D and 3D only.
- Scale: Enters via the mass parameter $m^2 > 0$ in the Lagrangian. In 2D and 3D the renormalization is super-renormalizable, scale is well-defined.

- 4D obstruction: ϕ_4^4 is not super-renormalizable; the continuum limit has been argued to be trivial (Aizenman 1982, Fröhlich 1982 — for ϕ_4^4 ; rigorous for $d > 4$, conjectural at $d = 4$). Yang-Mills is harder still.
- Wightman status in 4D: W1-W5 cannot be verified — construction does not extend.
- Reference: Glimm & Jaffe, *Quantum Physics: A Functional Integral Point of View* (1987).

A.2 Magnen-Rivasseau-Sénéor / Brydges-Federbush cluster expansion

- Achievement: Schwinger functions in 2D and 3D via convergent cluster expansion. Mass gap obtainable in finite volume; thermodynamic limit controllable in 2D/3D.
- Dimension: 2D, 3D.
- Scale: Cluster size parameter L , related to UV cutoff. Continuum limit $L \rightarrow 0$ controlled by power-counting renormalizability.
- 4D obstruction: Power counting fails for non-abelian gauge theory at zero coupling. Cluster expansion does not converge.
- Wightman status in 4D: W3 specifically open — mass gap not extractable.
- Reference: Brydges, Fröhlich & Spencer (1983); Magnen & Sénéor (1976).

A.3 Balaban multi-scale renormalization (1984-1989)

- Achievement: Ultraviolet stability of YM_4 on the unit torus $\mathbb{T}^4$ with gauge group $SU(2)$. Proves the partition function is finite and renormalizable. Multi-scale block-spin renormalization group analysis.
- Dimension: 4D, but only on $\mathbb{T}^4$ (finite volume).
- Scale: Lattice spacing $a \rightarrow$ continuum, and box size $L = 1 \rightarrow$ infinite. UV completion handled; IR completion not.
- 4D obstruction: The infinite-volume limit $\mathbb{T}^4 \rightarrow \mathbb{R}^4$ is open. Mass gap in finite volume not extracted because the torus has $\lambda_1(\mathbb{T}^4) > 0$ purely from compactness — this gap may not persist as $L \rightarrow \infty$.
- Wightman status: W1-W2 ok on $\mathbb{T}^4$. W3 in the $L \rightarrow \infty$ limit unverified.
- Reference: Balaban, *Comm. Math. Phys.* (1984-1989, series of papers).

A.4 Magnen-Rivasseau-Vauthier gauge-fixed constructive YM

- Achievement: Gauge fixing for constructive YM with explicit Slavnov-Taylor identities. Partial control of finite-volume YM_4 .
- Dimension: 4D in finite volume.
- Scale: Gauge parameter ξ enters explicitly; physical observables independent of ξ but the construction depends on it.
- 4D obstruction: Same as Balaban — infinite-volume limit not controlled.
- Wightman status: W3 open.
- Reference: Magnen, Rivasseau & Sénéor, “Construction of $Y-M_4$ with an infrared cutoff” *Comm. Math. Phys.* 155 (1993).

B. Lattice QCD

B.1 Wilson 1974 lattice gauge theory

- Achievement: Rigorous formulation of YM on a Euclidean lattice. Wilson action preserves gauge invariance exactly. Strong-coupling expansion shows confinement (area law for Wilson loops) and mass gap.
- Dimension: 4D on a lattice.
- Scale: Lattice spacing a . Continuum limit $a \rightarrow 0$ is the open problem.
- 4D obstruction: The continuum limit is not proved rigorously. Mass gap at finite a depends on bare coupling $g_0(a)$; the relation $\Lambda_{\text{QCD}} = a^{-1} \exp(-1/(2\beta_0 g_0^2))$ is a perturbative-RG relation that has not been proven to converge non-perturbatively to a continuum mass gap on $\mathbb{R}^4$.
- Wightman status: Lattice action is reflection-positive (Osterwalder-Seiler 1978), so W1-W5 hold on the lattice for any fixed $a > 0$. W4 (microcausality) is the difficult one in the continuum limit — Lorentz covariance is broken by the lattice.
- Reference: Wilson, *Phys. Rev. D* 10 (1974) 2445.

B.2 Creutz 1979 Monte Carlo

- Achievement: Numerical evidence for mass gap in lattice SU(2) gauge theory via Monte Carlo simulations. Demonstrates the cross-over from strong-coupling (confined) to weak-coupling (asymptotically free) regimes.
- Dimension: 4D lattice.
- Scale: Lattice spacing a and Monte Carlo step size.
- 4D obstruction: Numerical, not rigorous. Mass gap in lattice units, continuum extrapolation not controlled mathematically.
- Wightman status: Same as Wilson — W4 in continuum limit unverified.
- Reference: Creutz, *Phys. Rev. D* 21 (1980) 2308.

B.3 Morningstar-Pearson 1999 glueball spectrum

- Achievement: Precision lattice calculation of glueball masses for pure SU(3) YM. Mass $m(0^{++}) \approx 1.65\text{--}1.71$ GeV. Numerical mass gap confirmed.
- Dimension: 4D lattice.
- Scale: Lattice spacing, with multiple lattice spacings used for continuum extrapolation.
- 4D obstruction: Numerical extrapolation, not constructive. The continuum-limit Wightman QFT is not constructed; only correlators at finite spacing are computed.
- Wightman status: W3 numerically suggestive but not proven.
- Reference: Morningstar & Pearson, *Phys. Rev. D* 60 (1999) 034509; Chen et al., *Phys. Rev. D* 73 (2006) 014516.

B.4 Lüscher-Weisz improved actions

- Achievement: $O(a^2)$ -improved lattice actions reducing discretization artifacts. Precision spectroscopy possible.
- 4D obstruction: Same as Wilson — finite a only.
- Wightman status: Same as Wilson.
- Reference: Lüscher & Weisz, *Comm. Math. Phys.* 97 (1985) 59.

C. Perturbative QFT

C.1 Faddeev-Popov gauge fixing (1967)

- Achievement: Gauge-fixed perturbation theory for non-abelian gauge theories. Ghost fields introduced to cancel longitudinal gauge modes.
- Scale: Renormalization scale μ . Dimensional transmutation generates Λ_{QCD} via running coupling.
- 4D obstruction: Λ_{QCD} depends on the renormalization scheme; it is not directly the Wightman mass gap. Perturbative series is divergent (Dyson 1952) — even formal sum is not the actual mass gap.
- Wightman status: Perturbation theory does not construct a Wightman QFT. W3 cannot be addressed perturbatively.
- Reference: Faddeev & Popov, *Phys. Lett. B* 25 (1967) 29.

C.2 BRST quantization (1975)

- Achievement: Cohomological formulation of gauge symmetry via anticommuting ghost-antighost pairs. Slavnov-Taylor identities follow from BRST invariance.
- Scale: Same as Faddeev-Popov.
- 4D obstruction: Same as perturbation theory.
- Wightman status: BRST does not produce a Wightman theory; it organizes perturbation theory.
- Reference: Becchi, Rouet & Stora, *Ann. Phys.* 98 (1976) 287; Tyutin (1975, preprint).

C.3 'tHooft dimensional regularization + large N (1973-1974)

- Achievement: Dimensional regularization preserves gauge invariance. Large- N expansion for $SU(N)$ shows planar diagram dominance.
- Scale: Λ_{QCD} via dimensional transmutation. In large N , 't Hooft coupling $\lambda = g^2 N$ held fixed.
- 4D obstruction: Large- N limit is conjectured to be a string theory (via AdS/CFT for $\mathcal{N} = 4$ SYM); mass gap in $SU(N)$ pure YM at finite N remains conjectural.
- Wightman status: W3 not directly addressable from large- N alone.
- Reference: 't Hooft, *Nucl. Phys. B* 72 (1974) 461.

C.4 Wilsonian renormalization group (1971-1974)

- Achievement: Conceptual framework for understanding RG flow. Asymptotic freedom of YM coupling proved (Gross-Wilczek-Politzer 1973).
- Scale: RG scale μ , infrared cutoff Λ_{IR} , ultraviolet cutoff Λ_{UV} .
- 4D obstruction: Asymptotic freedom guarantees UV control but says nothing rigorous about the IR mass gap. The connection $\Lambda_{\text{QCD}} \sim \mu \exp(-1/(2\beta_0 g^2(\mu)))$ is a perturbative relation.
- Wightman status: RG is a framework, not a construction.
- Reference: Wilson, *Phys. Rev. B* 4 (1971) 3174-3205.

D. Stochastic / SPDE

D.1 Hairer regularity structures (2014)

- Achievement: Solution theory for singular SPDEs including ϕ_3^4 (the dynamic version). Fields-medal-level mathematical achievement.
- Dimension: 3D and below.
- Scale: Regularization scale, removed via renormalization.
- 4D obstruction: ϕ_4^4 regularity structure exists but the model is conjectured trivial (Aizenman-Fröhlich). Yang-Mills in 4D out of reach of current SPDE techniques.
- Wightman status: W3 in 4D — out of reach.
- Reference: Hairer, *Invent. Math.* 198 (2014) 269.

D.2 Gubinelli-Imkeller-Perkowski paracontrolled distributions (2013-)

- Achievement: Alternative SPDE framework. Φ_3^4 constructible.
- Dimension: $\leq 3\text{D}$.
- 4D obstruction: Same as Hairer for YM.
- Reference: Gubinelli, Imkeller & Perkowski, *Forum Math. Pi* 3 (2015) e6.

D.3 Chandra-Chevyrev-Hairer-Shen 2D YM via SPDE (2022-2024)

- Achievement: Stochastic quantization of 2D YM via Langevin dynamics. Existence of invariant measure proven.
- Dimension: 2D.
- 4D obstruction: 2D techniques do not extend to 4D — the noise regularity is too rough.
- Wightman status: W1-W5 obtainable in 2D, but 4D — open.
- Reference: Chandra, Chevyrev, Hairer & Shen, *Pub. Math. IHES* (2024).

E. AdS/CFT and Holographic Approaches

E.1 Maldacena AdS/CFT (1997)

- Achievement: Duality between $\mathcal{N} = 4$ super-Yang-Mills on $\mathbb{R}^4$ (boundary of AdS_5) and supergravity on $\text{AdS}_5 \times S^5$. Conjectured strong/weak coupling duality.

- Scale: AdS curvature radius L_{AdS} , related to 't Hooft coupling.
- 4D obstruction: Mass gap of pure $\text{SU}(N)$ YM (not $\mathcal{N} = 4$ SYM, which is conformal and massless) inferred from confining variations of AdS/CFT (Witten 1998). The construction is on AdS_5 background, not flat $\mathbb{R}^4$. The boundary $\mathbb{R}^4$ inherits its physics from the curved bulk — exactly the BST mechanism, but with a different curved bulk (AdS_5 vs D_{IV}^5).
- Wightman status: AdS/CFT is a conjectural duality, not a Wightman construction.
- Reference: Maldacena, *Adv. Theor. Math. Phys.* 2 (1998) 231.

E.2 Witten confinement via AdS (1998)

- Achievement: Argument for confinement in 4D YM via thermal AdS/CFT and Wilson-loop expectation values.
- Scale: AdS thermal radius (= compactification radius of Euclidean time).
- 4D obstruction: Conjectural; pure $\text{SU}(N)$ YM mass gap on $\mathbb{R}^4$ not rigorously derived.
- Wightman status: Conjectural.
- Reference: Witten, *Adv. Theor. Math. Phys.* 2 (1998) 505.

F. Other Approaches

F.1 Schwinger model (1962)

- Achievement: 2D QED with massless fermion. Analytically solvable. Mass gap from anomaly (axial anomaly generates photon mass).
- Dimension: 2D.
- 4D obstruction: 2D-specific. Does not generalize to 4D YM.
- Wightman status: 2D — verified.
- Reference: Schwinger, *Phys. Rev.* 128 (1962) 2425.

F.2 Polyakov confinement via instantons (1977)

- Achievement: Semiclassical confinement in 3D compact QED via monopole-instanton condensation. Mass gap from instanton sum.
- Dimension: 3D.
- Scale: Compactification radius / instanton size.
- 4D obstruction: 3D-specific (Polyakov's mechanism uses 3D-specific gauge theory features). 4D YM has instantons but the analogous argument doesn't close.
- Reference: Polyakov, *Nucl. Phys. B* 120 (1977) 429.

F.3 Stochastic quantization (Parisi-Wu 1981)

- Achievement: Quantum field theory as equilibrium of stochastic dynamics in a fictitious “5th time.”

- 4D obstruction: 4D YM stochastic quantization formal; rigorous construction not extended.
- Reference: Parisi & Wu, *Sci. Sinica* 24 (1981) 483.

F.4 Functional methods (Schwinger-Dyson)

- Achievement: Infinite tower of integral equations for n-point functions. Truncation schemes give approximate non-perturbative information.
- Scale: Truncation cutoff.
- 4D obstruction: Truncations are uncontrolled approximations, not rigorous constructions.
- Wightman status: Not a Wightman construction.
- Reference: Roberts & Williams, *Prog. Part. Nucl. Phys.* 33 (1994) 477.

F.5 Twistor methods (Penrose, Witten)

- Achievement: Reformulation of scattering amplitudes in twistor space; tree-level YM amplitudes computable.
- 4D obstruction: Perturbative tree-level only; does not address mass gap.
- Wightman status: N/A.
- Reference: Witten, *Comm. Math. Phys.* 252 (2004) 189.

Summary Pattern

Approach	Achievement	Where scale enters	W3 status on $\mathbb{R}^4$
Glimm-Jaffe	ϕ_2^4, ϕ_3^4	Mass parameter	Open (4D)
Brydges-Federbush	Cluster 2D/3D	Cluster size	Open (4D)
Balaban	YM_4 on $\mathbb{T}^4$	Compact box $L = 1$	Open ($L \rightarrow \infty$)
Magnen-Rivasseau-Vauthier	Finite-vol YM_4	Gauge param ξ	Open
Wilson lattice	Lattice YM_4	Lattice spacing a	Open ($a \rightarrow 0$)
Creutz Monte Carlo	Numerical mass gap	Lattice + MC	Numerical only
Morningstar-Glueball	Lattice spacing spectrum	Lattice spacing	Numerical only
Peardon	Improved lattice	Lattice spacing	Open
Lüscher-Weisz	Perturbation theory	Renormalization μ	Not addressable
Faddeev-Popov			

Approach	Achievement	Where scale enters	W3 status on $\mathbb{R}^4$
BRST	Cohomological gauge	RG scale μ	Not addressable
't Hooft large- N	Planar dominance	't Hooft coupling	Conjectural
Wilson RG	Asymptotic freedom	UV/IR cutoffs	Not addressable
Hairer SPDE	ϕ_3^4 regularity	Renormalization	Open (4D)
Gubinelli-Imkeller-Perkowski	Paracontrolled 3D	Cutoff	Open (4D)
Chandra-Chevyrev-Hairer-Shen	2D YM SPDE	Noise regularity	Open (4D)
Maldacena AdS/CFT	$\mathcal{N} = 4$ SYM duality	AdS radius L_{AdS}	Conjectural (curved bulk)
Witten thermal AdS	Confinement from AdS	Thermal radius	Conjectural
Schwinger model	2D QED mass gap	Anomaly scale	2D-specific
Polyakov instantons	3D compact QED	Compactification	3D-specific
Parisi-Wu stochastic	Formal 5th-time	Stochastic cutoff	Not extended
Schwinger-Dyson	Truncated tower	Truncation cutoff	Not rigorous
Twistor	Tree-level YM	N/A	Perturbative only

Twenty-two approaches across six schools. Zero deliver a Wightman QFT with mass gap on $\mathbb{R}^4$.

The Universal Pattern

Three categories of failure modes appear across the catalog:

- (1) Scale-from-cutoff: Lattice a , cluster size, box L , RG cutoff. The scale exists *because of the cutoff*. Removing the cutoff (continuum / thermodynamic limit) requires controlling the limit — open in 4D.
- (2) Scale-from-dimensional-transmutation: Λ_{QCD} from running coupling. This

scale is scheme-dependent and not derivable from the Wightman axioms alone. It is a *property of the renormalized perturbation theory*, not a property of an axiomatic QFT.

(3) Scale-from-curved-background: AdS_5 radius (AdS/CFT), torus $\mathbb{T}^4$ (Balaban), compactification (Polyakov). The scale exists *because the background is not $\mathbb{R}^4$* . Removing this restriction (going to flat $\mathbb{R}^4$) erases the scale and the gap.

The Curvature/Boundary Necessity claim is exactly the observation that category (3) is the only one that produces rigorous results, and only when the curved background is retained. BST's D_{IV}^5 is the BST-unique instance of category (3) that delivers all Wightman axioms with proven mass gap.

Implications for Paper YM-C

The catalog establishes the empirical content of the conjecture:

Conjecture (Curvature/Boundary Necessity, interacting case). *No Wightman QFT with positive mass gap exists on a complete, simply-connected, non-compact Riemannian manifold without boundary and with non-negative Ricci curvature.*

The 22 approaches above all fail in the same way: they require either compactness, boundary, curved bulk, or perturbative scale-smuggling. No approach succeeds purely on flat $\mathbb{R}^4$ with controlled limits.

This is empirical (not a proof), but it is the strongest evidence available after 50 years. Combined with the rigorous theorem of Paper YM-C Section 2 (Spectral Gap Necessity for the spectral-geometric case), it gives the conjecture overwhelming support.

Author note (Cal)

The catalog avoids over-claiming. Each entry states what the approach *achieves* and what it *does not* — not what it “can’t do in principle.” A future approach might succeed where these have not. But the structural pattern — every successful construction requires non-trivial geometry — is the substantive evidence.

This pattern is what Paper YM-C uses to motivate the Curvature/Boundary Necessity Theorem in Section 2, and what BST's D_{IV}^5 construction (Paper YM-B) demonstrates affirmatively: physics with mass gap lives on curved bounded geometries, not on flat infinite ones.

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